INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 15

1. Experiment Title:

Indexes and Adding a Row

2. Aim:

Create an index on the city column, find customers in Delhi, and insert a new customer.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

An index is a database object that can help the DBMS find rows more efficiently. CREATE INDEX creates an index on one or more columns. INSERT adds a new row to a table.

5. Algorithm / Procedure:

1. Create customer with customer_id as primary key and insert the four setup customers.
2. Create the idx_city index on the city column.
3. Select customers whose city is Delhi and sort by name.
4. Insert customer 5, Arjun, from Pune, then display the table.

6. Program / Queries / Commands:

```
CREATE TABLE customer (
    customer_id INT PRIMARY KEY,
    name VARCHAR(30) NOT NULL,
    city VARCHAR(20)
);

INSERT INTO customer (customer_id, name, city)
VALUES
    (1, 'Kiran', 'Kochi'),
    (2, 'Deepak', 'Delhi'),
    (3, 'Lata', 'Mumbai'),
    (4, 'Asha', 'Delhi');

-- Task 1
CREATE INDEX idx_city
ON customer (city);

-- Task 2
SELECT name FROM customer
WHERE city = 'Delhi'
ORDER BY name;

-- Task 3
INSERT INTO customer (customer_id, name, city)
VALUES (5, 'Arjun', 'Pune');

SELECT * FROM customer
ORDER BY customer_id;
```

7. Input:

Setup customers: Kiran/Kochi, Deepak/Delhi, Lata/Mumbai, Asha/Delhi. New customer: 5/Arjun/Pune.

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Task 2: name		
Asha		
Deepak		
Task 3: customer_id	name	city
1	Kiran	Kochi
2	Deepak	Delhi
3	Lata	Mumbai
4	Asha	Delhi
5	Arjun	Pune

9. Result:

The SQL statements for Experiment 15 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is an index in DBMS?
2. Which command creates an index?
3. What is the purpose of a primary key?
4. What does NOT NULL mean?